

LLM Foundations

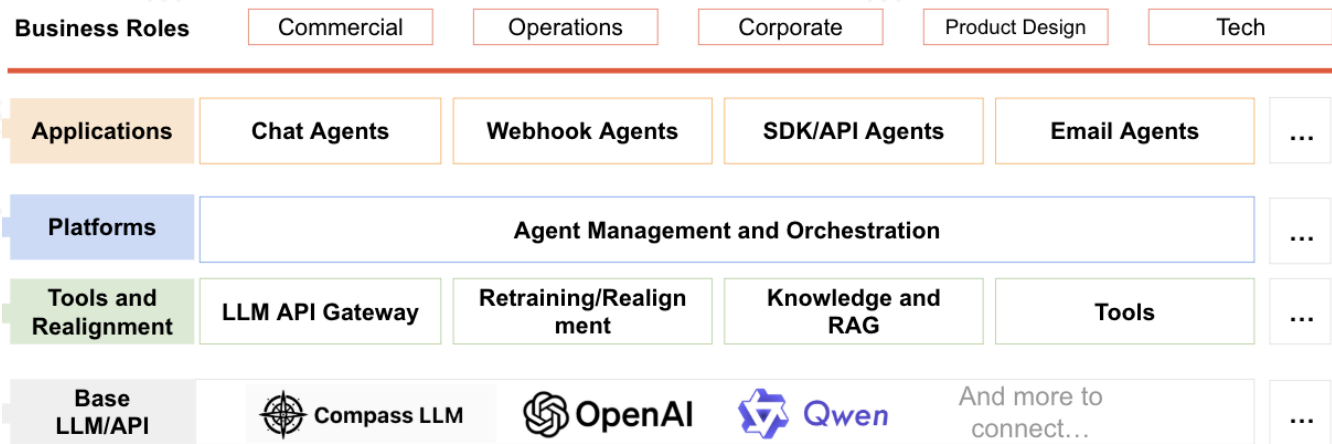
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1. How do different layers of LLM Framework work together to serve diverse business functions?

Business users interact with Applications (agents) that are built on top of a Platform layer for orchestration.

Supporting tools like Retraining, RAG, and API Gateways ensure the agents stay relevant and effective.

At the core, various LLM backends (OpenAI, Qwen, Compass) provide intelligence.



2. What is Layer Breakdown and what does each layer mean?

2.1. Applications Layer

Description: This is where business-facing agents live. Each agent is a functional unit that solves a task.

Application Agent Types:

- Chat Agents: LLM-powered chatbots for internal or external use
- Webhook Agents: Agents that listen for incoming triggers (e.g., user submissions, form input)
- SDK/API Agents: Agents embedded in apps or services via code/software integration
- Email Agents: Agents that generate/respond to emails, summarize threads, or triage

2.2. Platforms Layer (Smart)

Description: Used for Agent Management and Orchestration: This is the control center for building, connecting, and running agents.

Key capabilities:

- Agent lifecycle management: Manage how agents are created, configured, updated, paused, or decommissioned.
Example: users can version agents or disable outdated ones.
- Workflow orchestration: Define how agents interact in a sequence or in parallel to complete a larger task.
Example: Summarizer → Translator → Email Agent.
- Dependency tracking: Monitor which agents depend on outputs or decisions from others.
Example: Agent B waits for Agent A's result before proceeding.
- Trigger/event routing: Set up how agents are activated—based on user input, schedule, API call, or another agent's output.
Example: When a new document is uploaded, start the analysis workflow.

It enables users to coordinate multiple agents in a structured pipeline.

2.3. Tools and Realignment Layer

Description: This layer supports agent improvement, specialization, and alignment with business needs.

Functions of Tools and Realignment Layer:

- LLM API Gateway: Acts as a central access point to different large language model providers (e.g., OpenAI, Google Cloud Vertex AI, Amazon Web Services (AWS) Amazon Comprehend). Helps with load balancing, switching models, or testing alternatives.
 - Example of LLM API Gateway: Compass API Gateway efficiently routes the user's request to the appropriate microservice, ensuring that the text translation is performed accurately and securely
- Retraining/Realignment: Allows users to fine-tune or adjust agents based on new data or user feedback. This helps improve accuracy and personalization over time.
 - Example of Retraining/Realignment: After a user reports an irrelevant product logic response, the product assistant agent is retrained to deliver more accurate and aligned answers
- Knowledge and RAG: Integrates with internal knowledge bases using Retrieval-Augmented Generation (RAG). This lets agents access relevant documents or facts before answering, improving reliability.
 - Example of Knowledge and RAG: An RM assistant agent retrieves new seller policy references from an internal RM document repository to provide updated and compliant guidance during onboarding
- Tools: Expands the capabilities of AI agents beyond the knowledge of their original dataset by dynamically interacting with external resources and applications, real-time data, or other computational resources such as searching the web or retrieving data from an external database
 - Example of Tools: An online marketing agent uses a tool (Python Script) to automatically pull ad performance data from multiple channels i.e Google/ Apple/ Meta to detect abnormal campaign behavior, and suggest optimizations

2.4. Base LLM/API Layer

Description: This is the foundation of users agents. The LLMs and APIs that power everything above.

LLM Providers include:

- Compass LLM: Our in-house LLM. Used for:
 - Proprietary logic automation
 - Internal document handling (company-specific vocabulary)
 - Compliance or finance data extraction
 - Custom workflows requiring high control over data processing
 - **We only open it to whitelisted users, if you need it, please contact our dod**
- OpenAI: Industry-standard LLM provider (ChatGPT, GPT-4). Used for:
 - General-purpose chatbots
 - Email generation & summarization
 - Multistep reasoning agents (e.g., itinerary planner)
 - Report writing, legal draft review
- Qwen: Alibaba's open-source model, useful for multilingual or local context. Used for:
 - Multilingual chat or content translation (esp. Chinese, Southeast Asian languages)
 - Region-specific customer support agents
 - Localized search/retrieval tasks
- Gemini: Google/DeepMind's LLM. Used for:
 - Longcontext reasoning
 - Complex document analysis
- Others

3. What LLM Models supported on Smart?

3.1. OpenAI

Name	If support tools	Price Tag	Context Window	Multimodal	Reasoning Model	Input Cost per 1M tokens	Output Cost per 1M tokens
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GPT-5	Yes	Premium	400K	image	Reasoning Model	\$1.25 Cache: \$0.125	\$10.00
GPT-5 mini	Yes	Economical	400K	image	Reasoning Model	\$0.25 Cache: \$0.025	\$2.00
GPT-5 nano	Yes	Economical	400K	image	Reasoning Model	\$0.05 Cache: \$0.005	\$0.40
GPT-4.1 mini	Yes	Economical	1M	image		\$0.40 Cache: \$0.10	\$1.60
GPT-4o mini [Default]	Yes	Economical	128K	image		\$0.15 Cache: \$0.075	\$0.60
o4-mini	Yes	Premium	200K	image	Reasoning Model	\$1.10 Cache: \$0.55	\$4.40
o3-mini	Yes	Premium	200K	image	Reasoning Model	\$1.10 Cache: \$0.55	\$4.40
GPT-4o	Yes	Premium	128K	image		\$2.50 Cache: \$1.25	\$10.00
GPT-4.1	Yes	Premium	1M	image		\$2.00 Cache: \$0.50	\$8.00
GPT-4.1 nano	Yes	Economical	1M	image		\$0.10 Cache: \$0.025	\$0.40

3.2. Qwen

Name	If support tools	Price Tag	Context Window	Multimodal	Reasoning Model	Input Cost per 1M tokens	Output Cost per 1M tokens
QwQ-32B	No	Free	131K	/	Reasoning Model	Free	Free
Qwen2.5-32B-Instruct	Yes	Free	131K	/		Free	Free
Qwen2.5-VL-7B-Instruct	Yes	Free	32K	image		Free	Free
Qwen2.5-VL-72B-Instruct	Yes	Free	32K	image		Free	Free
Qwen3-Next-80B-A3B-Instruct	No	Economical	256K	/	No	\$0.15	\$1.20
Qwen3-Next-80B-A3B-Thinking	No	Economical	256K	/	Yes	\$0.15	\$1.20
Qwen3-Coder-480B-A35B-Instruct	Yes	Premium	256K	/	No	\$1.00	\$4.00
Qwen3-235B-A22B-Instruct	Yes	Economical	256K	/	No	\$0.25	\$1.00

3.3. Gemini

Name	If support tools	Price Tag	Context Window	Multimodal	Reasoning Model	Strength	Best Use Case	Input Cost per 1M tokens	Output Cost per 1M tokens	Link
Gemini 2.5 Pro	Yes	Premium	1M	image		Deep multimodal reasoning, long-context analysis	Complex document analysis, research assistant	\$1.25 \$2.5 > 200K input tokens	\$10 \$15 > 200K input tokens	https://ai.google.dev/gemini-api/docs/models#gemini-2.5-pro-preview-06-05
Gemini 2.5 Flash	Yes	Economical	1M	image		Cost-effective hybrid reasoning, optimized for high-volume, low-latency tasks	General chatbots, mass document parsing	\$0.3	\$2.5	https://ai.google.dev/gemini-api/docs/models#gemini-2.5-flash-preview

3.4. Claude

Name	If support tools	Price Tag	Context Window	Multimodal	Reasoning Model	Input Cost per 1M tokens	Output Cost per 1M tokens
Claude 3 Haiku	Yes	Economical	200k	image		\$0.25	\$1.25
Claude 3.5 Haiku	Yes	Economical	200k	image		\$0.80	\$4.00

Claude 4.5 Haiku	Yes	Economical	200k	image	Reasoning Model	\$1.00	\$5.00
Claude 3.7 Sonnet	Yes	Premium	200k	image	Reasoning Model	\$3.00	\$15.00
Claude 4 Sonnet	Yes	Premium	200k	image	Reasoning Model	\$3.00	\$15.00
Claude 4.5 Sonnet	Yes	Premium	200k	image	Reasoning Model	\$3.00	\$15.00

3.5. DeepSeek

Name	If support tools	Price Tag	Context Window	Multimodal	Reasoning Model	Input Cost per 1M tokens	Output Cost per 1M tokens
DeepSeek-V3.1	Yes	Economical	160K	/	Yes	\$0.6	\$1.7
DeepSeek-R1	Yes	Premium	160K	/	Yes	\$1.35	\$5.4